



Advanced Cisco Networking Academy – Layer 2 Attacks and Mitigation

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Purpose

Securing Layer 2 vulnerabilities is an increasingly vital skill as our society starts to offload all information off paper, and onto the internet. Without security these vulnerabilities companies are at risk of leaking sensitive information regarding their products and services, and leaking customer information, leading to customers suing for damages.

This lab helps to understand how to secure layer 2 vulnerabilities by exploiting them, giving further insight as to how and why certain attacks work. This understanding of how vulnerabilities are exploited and why attacks work develops a skillset beyond the mitigation of specific attacks done in this lab, but all Layer 2 attacks that exploit the same mechanism that creates vulnerabilities.

Background

There are multiple layers in a network, starting from layer one and ending at layer 7. The first layer is the physical layer; this is the wiring between devices and the devices that make up the network themselves. The focus of this lab is the second layer; this layer is comprised of MAC-addresses, and physical addresses. This layer manages node to node exchange, meaning end-device to end-device (e.g. phone to phone). This layer oversees any routing of packets within the same network. Once the packet leaves through the router, it becomes layer 3. In intranet work communications it is important to know that most traffic does not go to router but is routed through the switch as layer 3 communication uses IP addresses that are only needed for communication outside of the network.

A switch is a layer 2 device, responsible for routing within the network. The main function of a switch is to keep track of what MAC addresses come from which port. To do this, the switch keeps a MAC address-table listing every MAC address with the port number it came from. This is the exact mechanism that is exploited multiple times in this lab. An unconfigured switch will allow any host connected to the switch to broadcast real, fake, or a stolen MAC address and have the switch bind that address to the port the host is connected to. Using this mechanism attackers can steal data meant for another user, overflow the table so that the switch crashes, and many more Man in The Middle and Denial of Service attacks.

One of the attacks done in this lab is known as the ARP spoofing or ARP poisoning attack. This attack exploits the fact that the ARP protocol itself is trusted by all network devices by default, meaning that any fake packets are trusted, and ARP tables are updated accordingly based off the forged packets. ARP or Address Resolution Protocol is a layer 2 protocol used to map MAC addresses to IP addresses within the same network. Every device in the network (Phones, Computers, Switches,

Routers) typically keeps an ARP table; this enables the network devices to keep track of who has what IP and MAC address. This attack occurs when the attacker sends ARP packets to the victim saying that the attackers' MAC-address is bound to the router's/default gateway's IP address and telling the default gateway that the attackers' MAC is bound to the Victims IP. When the attack is successful, all packets that the victim sends to the gateway, and all packets in the gateway sends to the victim are first routed through the attacker's machine.

Typically, the goal of ARP Spoofing is not to deny service to the victim (although possible), but rather to capture and steal information that the victim sends to the gateway. Another thing that exploiters can do with ARP spoofing is modifying packets that come back from the internet that was intended for the Victim, attackers can eavesdrop for bank credentials, then create an identical website making the victim think their account has not problems, but by the time they realize that someone had drained their account it could be weeks, making them virtually untraceable. Besides the data theft and manipulation, this is also a way that Black Hatters inject malicious code into their victims and have that malware spread. Having no safeguards with respect to ARP spoofing attacks poses significant security risks to a company. The scale at which these attacks are carried out every day is also significant. SentinelOne, an enterprise grade cybersecurity platform, reports that more than 30,000 ARP spoofing attacks are carried out daily.

This significant risk that ARP spoofing attacks carry towards companies emphasizes the importance of enabling DAI or Dynamic ARP and DHCP Snooping Inspection to prevent ARP Spoofing attacks. DAI works by referencing the ARP packets with a trusted DHCP snooping table, effectively blocking any Gratuitous ARP packets, preventing the packets from reaching the victim completely. This works because of how DHCP tables work; each unique MAC address on a DHCP table is assigned to one IP address. When DAI is enabled and inspects the ARP packet, sees that the MAC address to IP address mapping differs from the already existing one on the DHCP snooping table, then it blocks that ARP packet. The switch creates a trusted DHCP snooping table through inspecting DHCP packets between the hosts and the DHCP server, effectively creating a DHCP snooping table to reference ARP packets.

This segways into another method of exploiting MAC-address bindings with something called a DHCP starvation attack. DHCP starvation is another attack done in this lab; it works by exploiting the limited number of IP addresses a DHCP server has available to lease. As mentioned previously DHCP servers work by giving every unique MAC-address a unique IP address in the network, this setup is usually effective in giving all host devices a unique private IP address to communicate with each other and to the router to access the internet. But if a single host decides to forge more MAC-

addresses than available and send packets to the DHCP server asking for an IP address for each MAC address through a DHCPDiscover packets, that single host can effectively deny service to all other hosts in the network as DHCP servers can only lease a limited number of addresses and the other hosts need an IP address to access the internet and other resources.

The method to prevent this attack is much simpler than DAI or DHCP snooping. The method to prevent DHCP starvation is by enabling port security on the switch connected to end devices. Port security is a configuration on the switch that restricts the maximum number of MAC-addresses allowed from a single port, and the behavior of the port when the maximum number of MAC-addresses from a port exceeds the allowed number of MAC-addresses. There are 3 different ways that a port can behave when the policy set on the port is violated; protect, which simply restricts all MAC-addresses which violates the set number allowed allowing legitimate traffic to pass through, this mode is effective when port shutdown is not an option for the network. The second mode is restricted does the same thing, but it generates a log of how many times the policy has been violated, and can be configured to send a report to a SNMP server that the policy has been violated, this is best when the port cannot be shut down but strict access control and monitoring is needed. The final mode, shutdown is most extreme. It does everything restrict but shuts down the port when the maximum number of MAC-addresses is exceeded; this is best when configuring a zero-trust environment.

Another similar attack is the CAM overflow attack. This attack is prevented in the same way as the DHCP starvation attack as it involves sending an immense number of packets with MAC addresses. As mentioned in the second paragraph, switches operate by keeping track of which MAC addresses come from which port. The switch has a table for this, but there is a limit to how many entries this table can have as the switch comes with a limited memory. When the attack sends a massive number of packets, claiming multiple MAC-addresses, the memory is overloaded and the switch can shut down causing service to go down in the network, making it a Denial-of-Service attack.

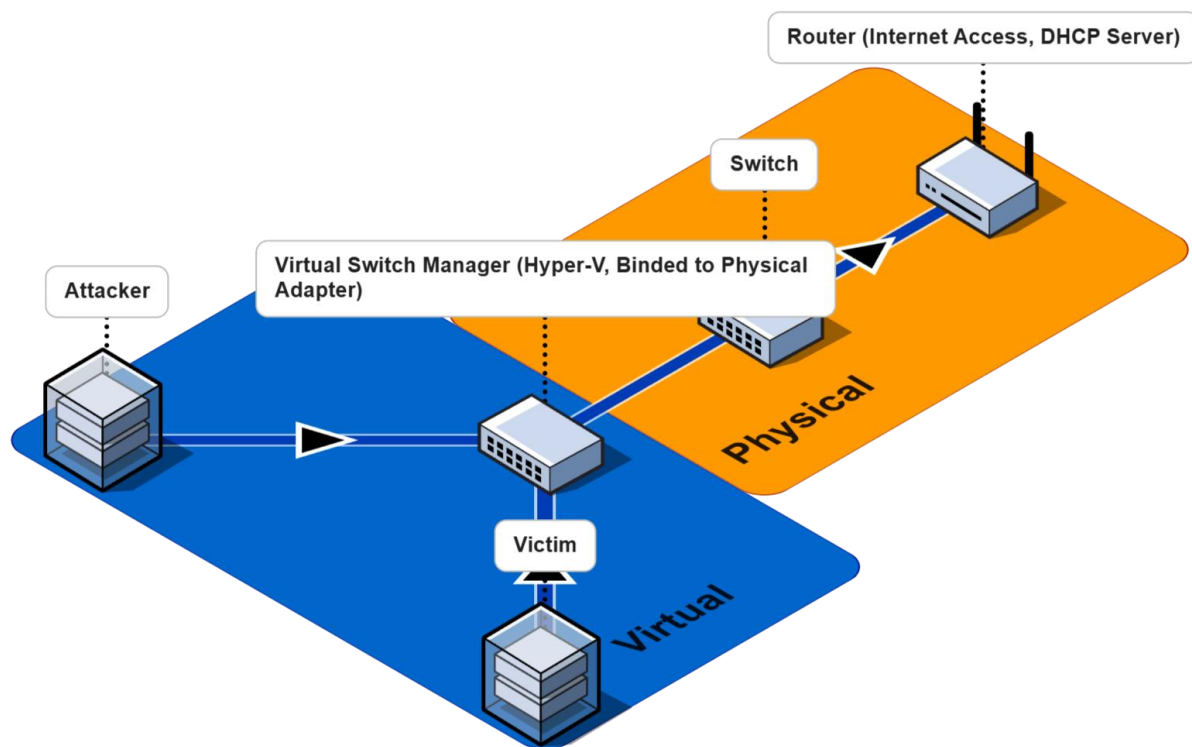
The three attacks mentioned in this background are all the Layer 2 attacks done in this lab. These are all categorized as Layer 2 due to how it exploits the nature of layer 2 protocols that use MAC addresses. Although there are only three attacks and the mitigations for those attacks in this lab there are many more layer 2 attacks such as STP manipulation where Spanning Tree Protocol is manipulated to intercept packets with a rogue switch, VLAN hopping attacks to gain access to otherwise unobtainable information within the same network, and many more attacks. These attacks emphasize the importance of having the knowledge of how to both mitigate

and perpetrate these attacks, to further develop an understanding of layer 2 operations and mitigation of Layer 2 attacks beyond the attacks mentioned in this lab.

Lab Summary

In this lab I performed ARP spoofing, DHCP starvation, and CAM overflow attack using two VMs connected to a physical switch connecting to a router connected to an ISP. To do this I created VMs using Hyper-V, a built-in virtualization software in Windows 11 Pro, Enterprise, or Education. Kali Linux operating system was used in virtual machines as it comes with hacking tools. Then, I used the tools in Kali to perform the attacks. To enable Kali VMs to update their repository, internet was needed, so the router was configured to provide internet along with a DHCP server to do the DHCP starvation attack.

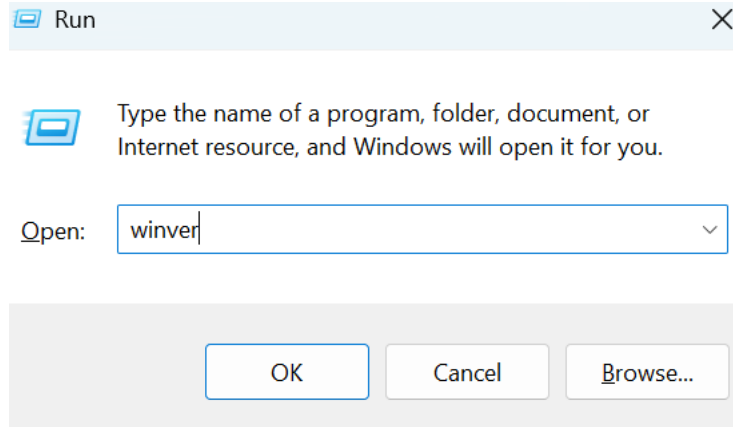
Network Diagram



Lab Commands

Setting up VMs

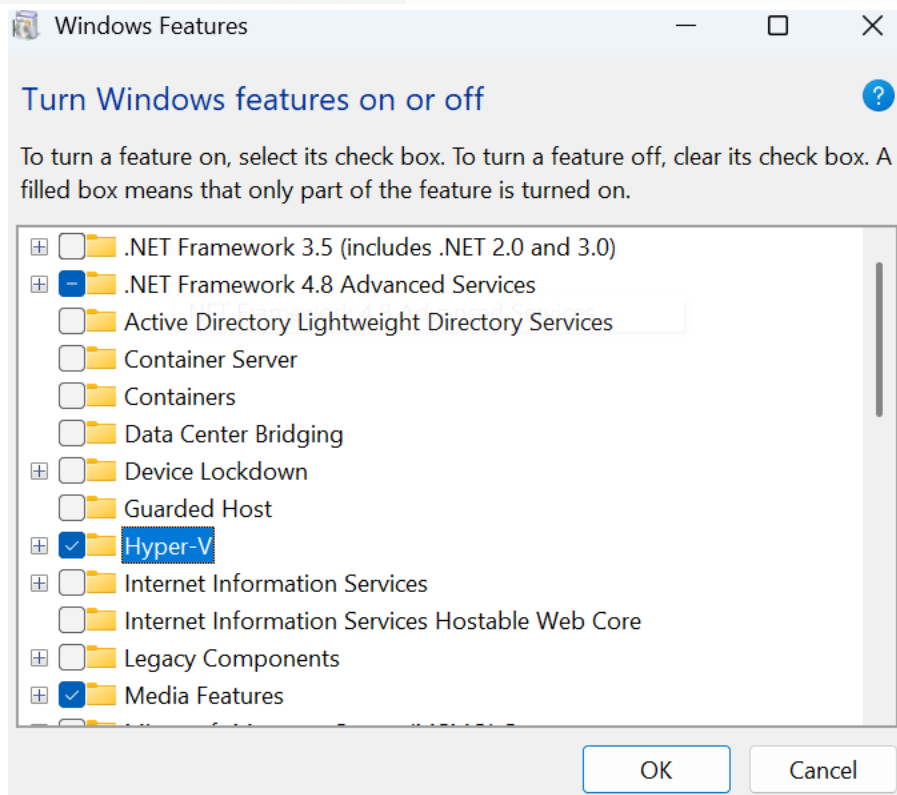
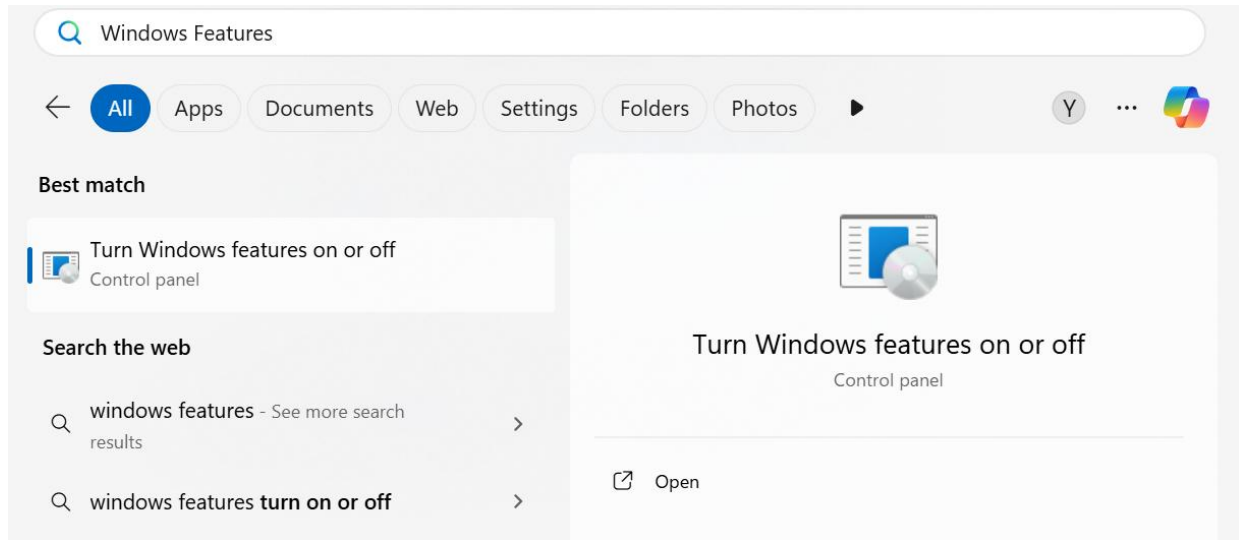
To set up the VMs using Hyper-V first click Windows + R and type in “winver”



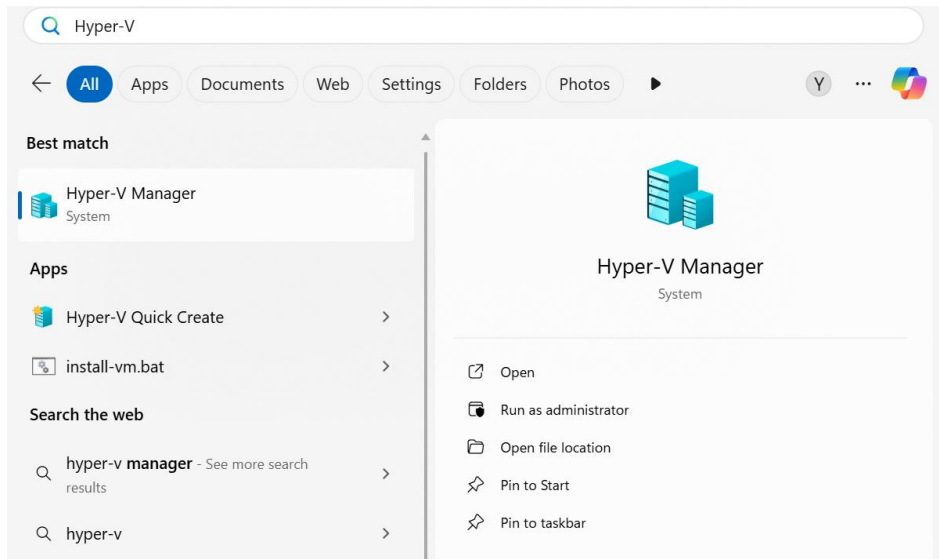
Windows version needs to be either Pro, Enterprise, or Education



Once this is confirmed, enable Hyper-V by pressing the windows key and typing “windows features” and running the first option and making sure the Folder with Hyper-V has a blue checkmark next to it.



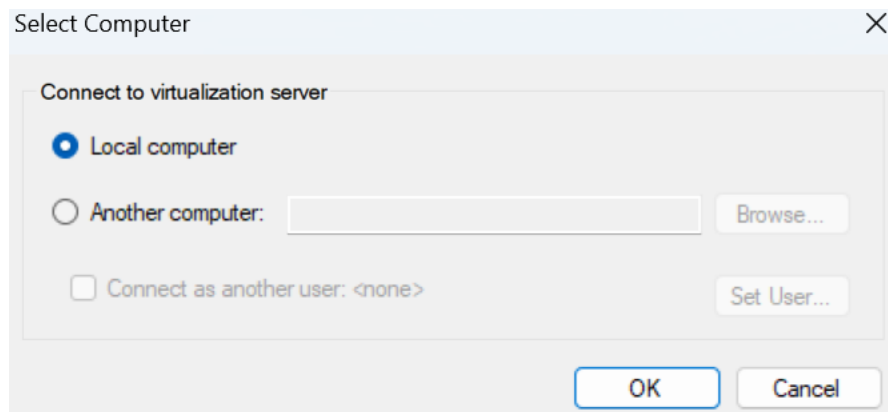
Then, press the windows key and type Hyper-V and click open



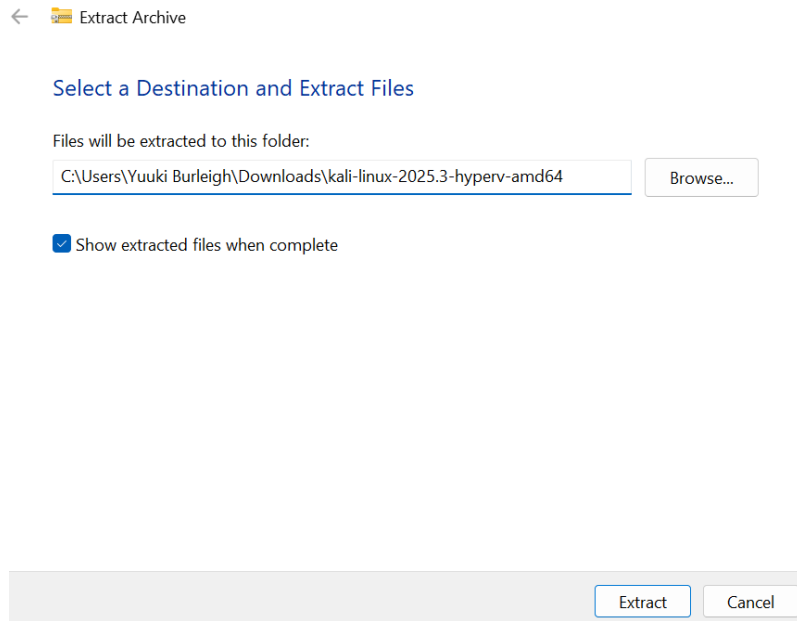
Once in Hyper-V click connect to server to start



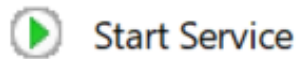
Choose local computer



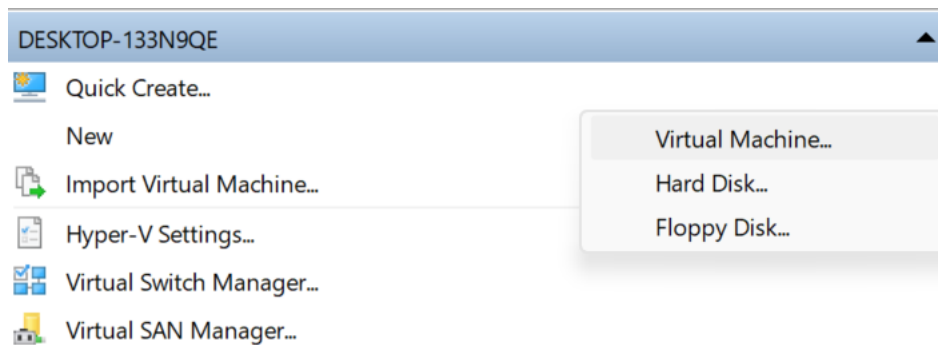
Now navigate to a browser and search for a Hyper-V virtual drive with Kali Linux or use this link: <https://www.kali.org/get-kali/#kali-virtual-machines> and download. Once the Zip Folder if downloaded extract it



Once extracted, navigate back to Hyper-V and hit start service



Click “New” and select “Virtual Machine”



Create a Name for the Machine

New Virtual Machine Wizard

Specify Name and Location

Before You Begin

Specify Name and Location

Specify Generation

Assign Memory

Configure Networking

Connect Virtual Hard Disk

Installation Options

Summary

Choose a name and location for this virtual machine.

The name is displayed in Hyper-V Manager. We recommend that you use a name that helps you easily identify this virtual machine, such as the name of the guest operating system or workload.

Name:

You can create a folder or use an existing folder to store the virtual machine. If you don't select a folder, the virtual machine is stored in the default folder configured for this server.

☐ Store the virtual machine in a different location

Location:

 If you plan to take checkpoints of this virtual machine, select a location that has enough free space. Checkpoints include virtual machine data and may require a large amount of space.

< Previous

Next >

Finish

Cancel

Opt for Gen 2 and Allow for Dynamic Memory so that machine can use as much memory as it needs.



Specify Generation

- Before You Begin
- Specify Name and Location
- Specify Generation**
- Assign Memory
- Configure Networking
- Connect Virtual Hard Disk
- Installation Options
- Summary

Choose the generation of this virtual machine.

☐ Generation 1

This virtual machine generation supports 32-bit and 64-bit guest operating systems and provides virtual hardware which has been available in all previous versions of Hyper-V.

☒ Generation 2

This virtual machine generation provides support for newer virtualization features, has UEFI-based firmware, and requires a supported 64-bit guest operating system.

 Once a virtual machine has been created, you cannot change its generation.

[More about virtual machine generation support](#)

< Previous

Next >

Finish

Cancel



Assign Memory

- Before You Begin
- Specify Name and Location
- Specify Generation
- Assign Memory**
- Configure Networking
- Connect Virtual Hard Disk
- Installation Options
- Summary

Specify the amount of memory to allocate to this virtual machine. You can specify an amount from 32 MB through 251658240 MB. To improve performance, specify more than the minimum amount recommended for the operating system.

Startup memory: MB

☒ Use Dynamic Memory for this virtual machine.

 When you decide how much memory to assign to a virtual machine, consider how you intend to use the virtual machine and the operating system that it will run.

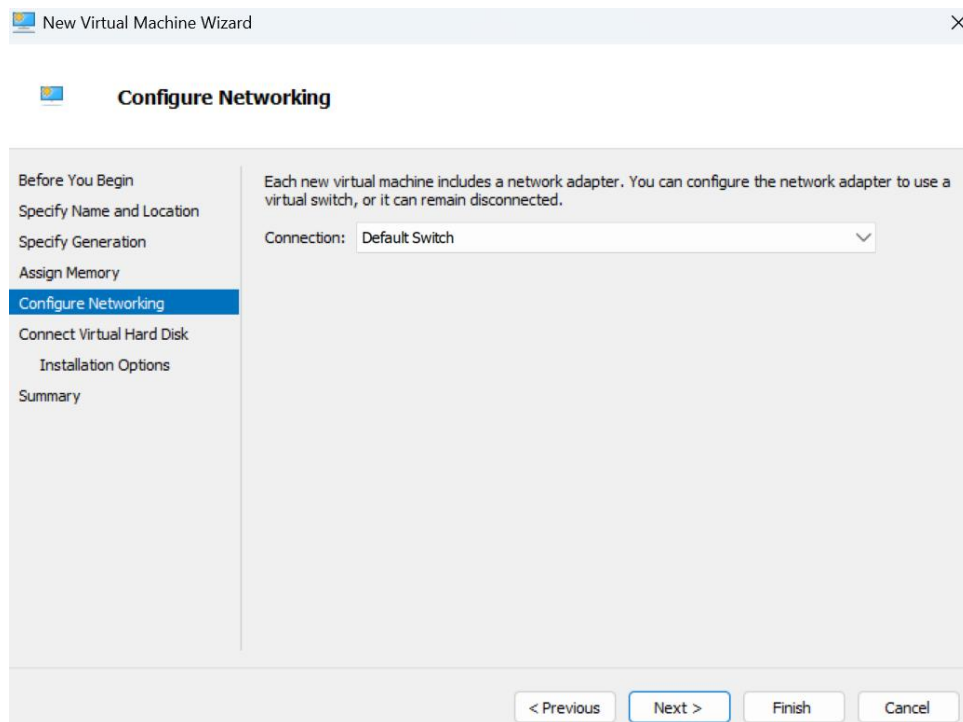
< Previous

Next >

Finish

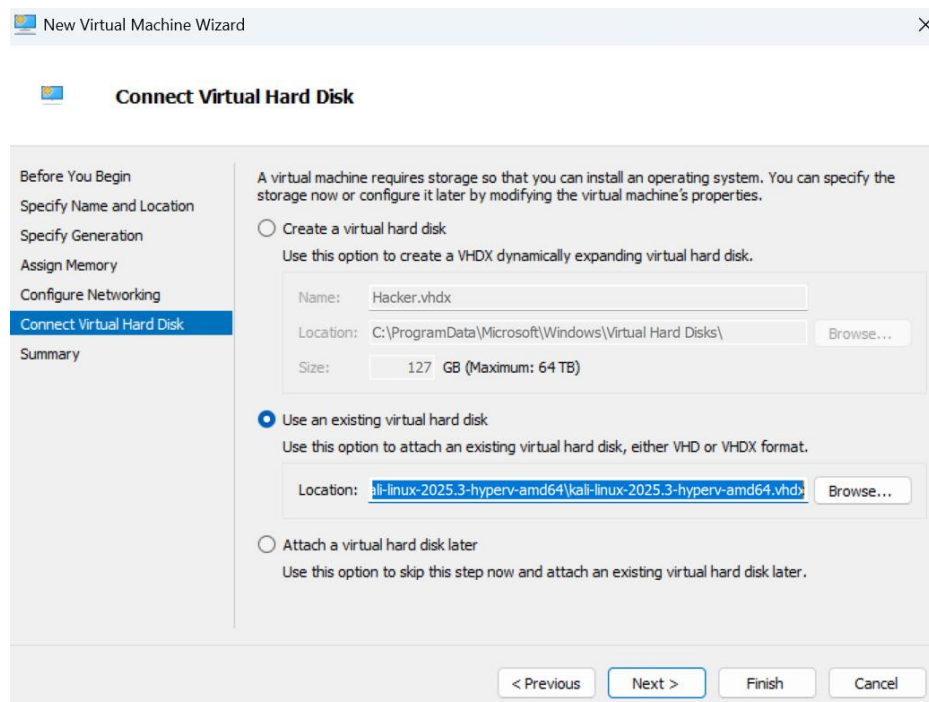
Cancel

This will be changed later but connect the machine to the default switch



The screenshot shows the 'Configure Networking' step of the 'New Virtual Machine Wizard'. The left sidebar lists the steps: 'Before You Begin', 'Specify Name and Location', 'Specify Generation', 'Assign Memory', 'Configure Networking' (highlighted), 'Connect Virtual Hard Disk', 'Installation Options', and 'Summary'. The main area contains the text: 'Each new virtual machine includes a network adapter. You can configure the network adapter to use a virtual switch, or it can remain disconnected.' Below this, the 'Connection:' dropdown menu is set to 'Default Switch'. At the bottom, there are four buttons: '< Previous', 'Next >', 'Finish', and 'Cancel'.

Now, opt to choose an existing virtual hard disk; this disk is located within the folder recently extracted.

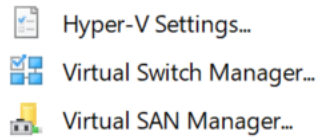


The screenshot shows the 'Connect Virtual Hard Disk' step of the 'New Virtual Machine Wizard'. The left sidebar lists the steps: 'Before You Begin', 'Specify Name and Location', 'Specify Generation', 'Assign Memory', 'Configure Networking', 'Connect Virtual Hard Disk' (highlighted), and 'Summary'. The main area contains the text: 'A virtual machine requires storage so that you can install an operating system. You can specify the storage now or configure it later by modifying the virtual machine's properties.' There are three radio button options: 'Create a virtual hard disk' (unselected), 'Use an existing virtual hard disk' (selected), and 'Attach a virtual hard disk later' (unselected). Under 'Create a virtual hard disk', it says 'Use this option to create a VHDX dynamically expanding virtual hard disk.' and shows fields for 'Name: Hacker.vhdx', 'Location: C:\ProgramData\Microsoft\Windows\Virtual Hard Disks\' (with a 'Browse...' button), and 'Size: 127 GB (Maximum: 64 TB)'. Under 'Use an existing virtual hard disk', it says 'Use this option to attach an existing virtual hard disk, either VHD or VHDX format.' and shows a 'Location:' field with the path 'kali-linux-2025.3-hyperv-amd64\kali-linux-2025.3-hyperv-amd64.vhdx' (highlighted) and a 'Browse...' button. At the bottom, there are four buttons: '< Previous', 'Next >', 'Finish', and 'Cancel'.

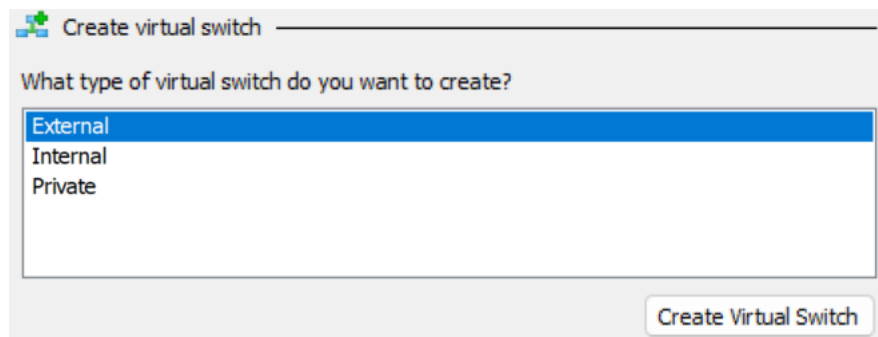
Click next and finish, copy and paste the virtual drive used in the file system to repeat this process for the victim's Virtual Machine.

Next create a Virtual Switch that is binded to the network adapter that will be used to connect to the physical network.

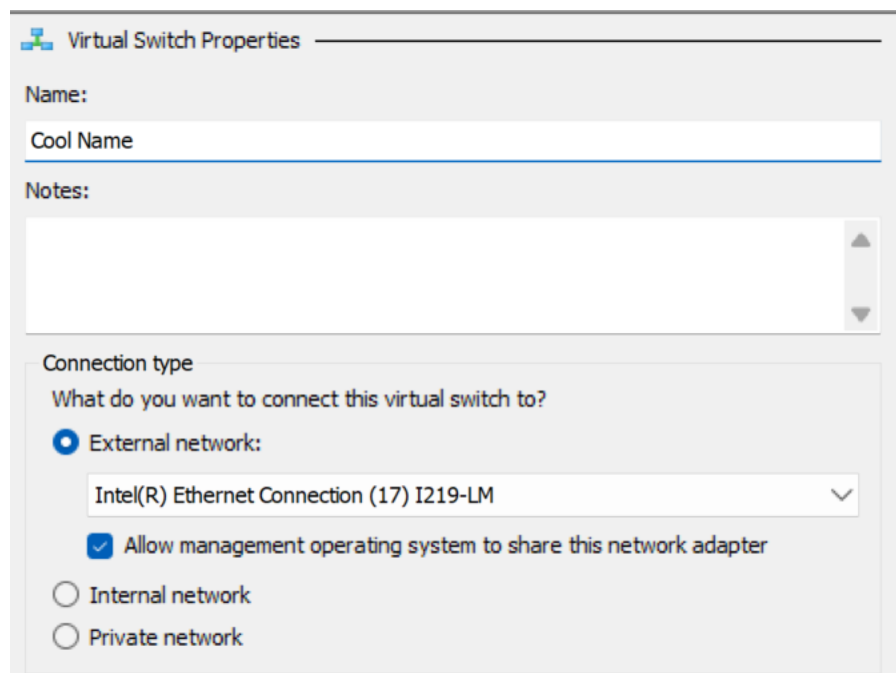
Click “Virtual Switch Manager”



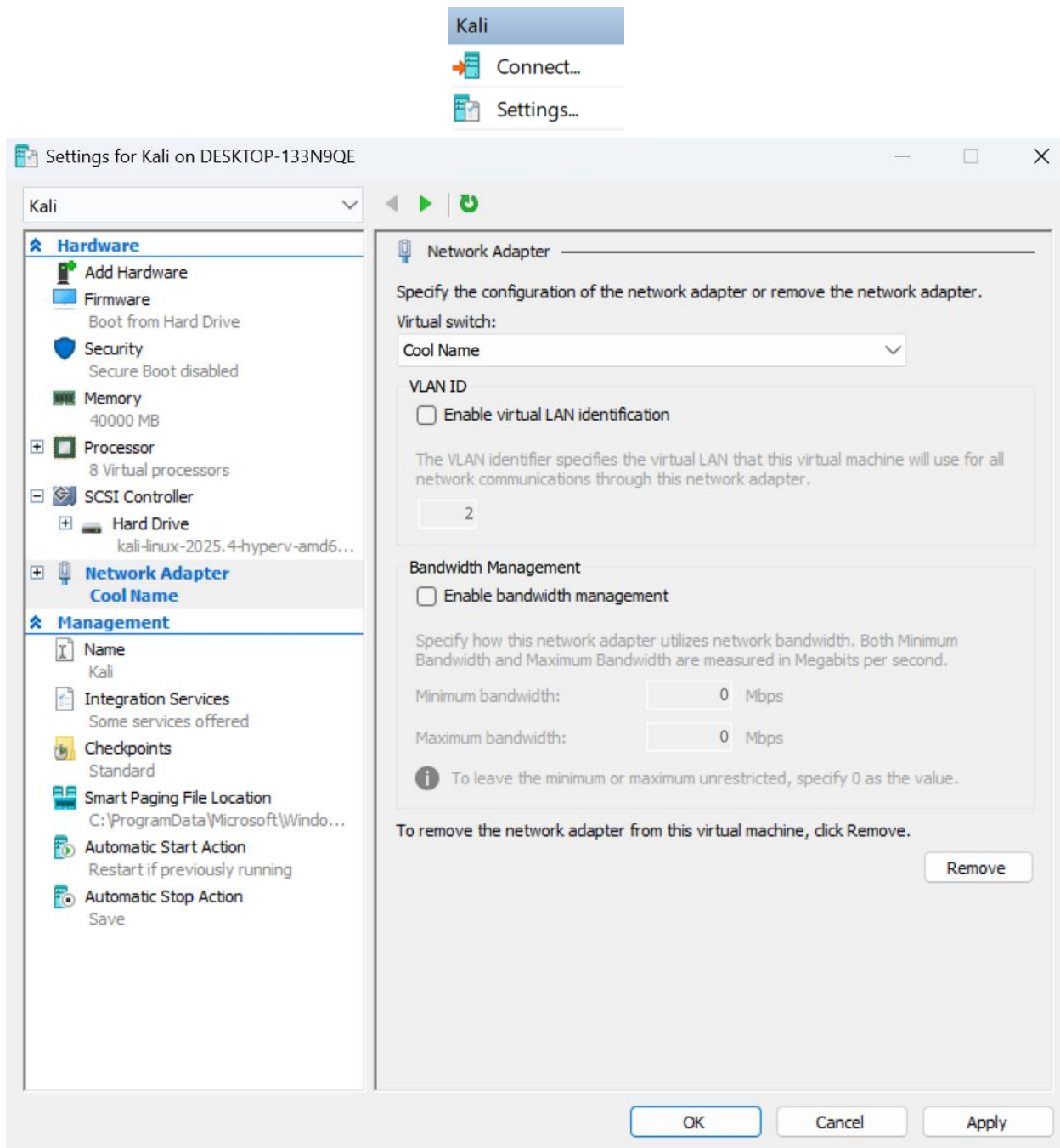
Choose to create an external switch, and click Create Virtual Switch



Assign a cool name to the switch and choose the network adapter to connect to the physical network and allow it to be shared.



Now go to the settings of both virtual machines and change the switch to the one with a cool name. This will allow the VMs to access the network connected through the adapter.



Click OK to confirm the changes
That concludes setting up the VMs

VM Commands

To access the terminal in Kali Linux, click the black box that looks like a terminal in the top left corner of the screen.

- **Sudo apt update** – this command updates all the repositories for the tools in Kali Linux this command is needed to run before executing a command
- **Ip a** – this is the command to find out what interface has what IP address typically the interface is eth0
- **Sudo macof -i [interface]** - this command is to run the MAC overflow attack, the -i is to specify an interface to send the packets out of
- **Sudo arpspoof -i [interface] [victim ip address]** - this command is to perform the ARP spoofing attack
- **Sudo yersinia -I** – this activates the tools used for DHCP starvation
 - To do this using Yersinia tap F2 on the keyboard to switch to DHCP attacks. Then, press X and choose the number to send DHCP to discover packets.

Switch Commands

- **Ip dhcp snooping vlan [#]** – this enables DHCP snooping on the specified VLAN
- **Ip arp inspection vlan [#]** – this enables DAI or dynamic ARP inspection
- **Ip arp inspection trust** – this command is done at the interface level to interfaces that are trusted, typically non-end user interfaces
- **Ip arp inspection limit rate [#]** – this limits the number of ARP packets allowed to be accepted into an interface and is done at the interface level
- **Switchport port-security** – done at the interface level and default max MAC-addresses allowed per port is 1 and violation mode is shutdown
- **Switchport port-security max [#]** – configures the max number of MAC-addresses allowed per port
- **Switchport port-security violation mode (protect/restrict/shutdown)** – this sets the port security violation mode

Proof of Attacks

Macof:

```
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo macof -i eth0
[sudo] password for kali:
32:a6:6d:7c:c0:12 0:53:4a:15:59:ab 0.0.0.0.19685 > 0.0.0.0.38129: S 303545990:303545990(0) win 512
ce:a0:4b:76:6d:9d f0:bb:3b:31:55:a 0.0.0.0.58301 > 0.0.0.0.1055: S 635836793:635836793(0) win 512
4f:7c:b5:68:c0:50 ee:6c:b1:74:34:e7 0.0.0.0.3824 > 0.0.0.0.34187: S 1279470369:1279470369(0) win 512
80:ca:7f:27:53:d1 b:79:f0:35:80:1d 0.0.0.0.9399 > 0.0.0.0.51602: S 1030985050:1030985050(0) win 512
b5:63:e2:2d:28:d6 aa:80:26:28:a6:1f 0.0.0.0.53918 > 0.0.0.0.50235: S 977333216:977333216(0) win 512
27:27:b1:3f:2f:68 d7:3c:a5:3:16:94 0.0.0.0.13386 > 0.0.0.0.24148: S 497581678:497581678(0) win 512
9f:c4:92:74:ee:48 bf:30:ad:21:aa:3d 0.0.0.0.15891 > 0.0.0.0.35622: S 1439332094:1439332094(0) win 512
63:6e:fc:6d:b3:da 96:a9:de:9:d:ef 0.0.0.0.54989 > 0.0.0.0.52471: S 1083461199:1083461199(0) win 512
e:b9:24:6:77:75 dd:2a:d9:14:ed:f5 0.0.0.0.61128 > 0.0.0.0.15831: S 1062918292:1062918292(0) win 512
43:9d:5d:1a:33:9d 25:b6:d5:77:3:ce 0.0.0.0.31599 > 0.0.0.0.6385: S 682862522:682862522(0) win 512
5f:ec:3c:4d:1d:16 20:cd:d0:8:f5:95 0.0.0.0.8201 > 0.0.0.0.56311: S 1293980785:1293980785(0) win 512
a5:9e:57:56:80:4d cf:e3:80:2f:82:c9 0.0.0.0.17346 > 0.0.0.0.12846: S 629812020:629812020(0) win 512
65:9f:c1:24:77:c8 8a:b9:84:10:8a:55 0.0.0.0.29214 > 0.0.0.0.4585: S 1712500058:1712500058(0) win 512
54:4:70:30:b9:99 f5:2f:c3:55:77:d1 0.0.0.0.39153 > 0.0.0.0.20407: S 1728778518:1728778518(0) win 512
f:be:4d:28:bb:af e2:7a:42:45:df:a1 0.0.0.0.36667 > 0.0.0.0.48384: S 342225015:342225015(0) win 512
57:d9:66:22:dc:8f e7:82:69:21:e1:92 0.0.0.0.10058 > 0.0.0.0.22404: S 1218980683:1218980683(0) win 512
5c:71:31:2b:a1:2f 5d:6e:43:3b:51:a1 0.0.0.0.54896 > 0.0.0.0.39433: S 1052831170:1052831170(0) win 512
6c:c7:32:49:d1:ab 42:97:f8:27:4e:42 0.0.0.0.151 > 0.0.0.0.56556: S 375115135:375115135(0) win 512
5b:97:56:5c:d6:a6 ea:80:54:7e:42:1a 0.0.0.0.13220 > 0.0.0.0.24834: S 1077001684:1077001684(0) win 512
a6:35:83:15:30:2f 90:8e:dc:5c:3:a4 0.0.0.0.16910 > 0.0.0.0.42737: S 1718299058:1718299058(0) win 512
77:8e:78:5a:1e:f1 a1:db:44:3c:b9:25 0.0.0.0.13881 > 0.0.0.0.13316: S 1979123262:1979123262(0) win 512
12:25:7:34:99:9d 8f:aa:1b:22:fc:a5 0.0.0.0.3148 > 0.0.0.0.60288: S 458835273:458835273(0) win 512
6f:4a:ee:4e:ef:7a 50:f9:3:32:ff:d8 0.0.0.0.26792 > 0.0.0.0.24832: S 238973122:238973122(0) win 512
d0:90:4e:6d:39:ff 79:83:9:51:71:6c 0.0.0.0.38339 > 0.0.0.0.23340: S 2113333994:2113333994(0) win 512
e4:5e:7e:27:fc:f 9b:39:fa:6c:73:9 0.0.0.0.60858 > 0.0.0.0.49732: S 674356471:674356471(0) win 512
64:8c:69:31:66:27 f2:ad:99:23:b4:85 0.0.0.0.20357 > 0.0.0.0.26894: S 1028992484:1028992484(0) win 512
69:fb:56:7f:b4:be 81:bf:be:78:e2:7e 0.0.0.0.13769 > 0.0.0.0.49385: S 571228064:571228064(0) win 512
eb:5c:99:13:84:9e b0:30:72:f:86:96 0.0.0.0.34813 > 0.0.0.0.38322: S 1586800877:1586800877(0) win 512
b6:c8:45:5d:51:35 5a:8c:ed:5c:a8:76 0.0.0.0.17293 > 0.0.0.0.2699: S 1573123533:1573123533(0) win 512
d3:ca:3c:5:36:f1 26:30:15:e:54:8a 0.0.0.0.28360 > 0.0.0.0.42466: S 1530469429:1530469429(0) win 512
95:c6:6c:6a:20:79 ec:4c:79:4d:45:f7 0.0.0.0.56814 > 0.0.0.0.38819: S 1260817047:1260817047(0) win 512
97:fc:ce:12:4d:53 2c:bc:3d:5c:f1:88 0.0.0.0.5553 > 0.0.0.0.3458: S 1155432894:1155432894(0) win 512
f4:7e:d9:32:91:4c b5:fc:73:2a:1a:af 0.0.0.0.2662 > 0.0.0.0.17837: S 1692930736:1692930736(0) win 512
15:f4:e7:2d:45:d9 bc:9f:46:6d:1:41 0.0.0.0.13798 > 0.0.0.0.44665: S 162416390:162416390(0) win 512
54:35:1e:3:9d:43 ca:13:92:4f:80:f1 0.0.0.0.26803 > 0.0.0.0.57920: S 1212541762:1212541762(0) win 512
be:ed:27:49:36:6a 2:37:ff:3d:73:ea 0.0.0.0.22337 > 0.0.0.0.9142: S 806310804:806310804(0) win 512
51:4a:bd:5d:a9:47 37:43:c8:7f:d:ea 0.0.0.0.39280 > 0.0.0.0.48784: S 2145043003:2145043003(0) win 512
```

With Mitigations:

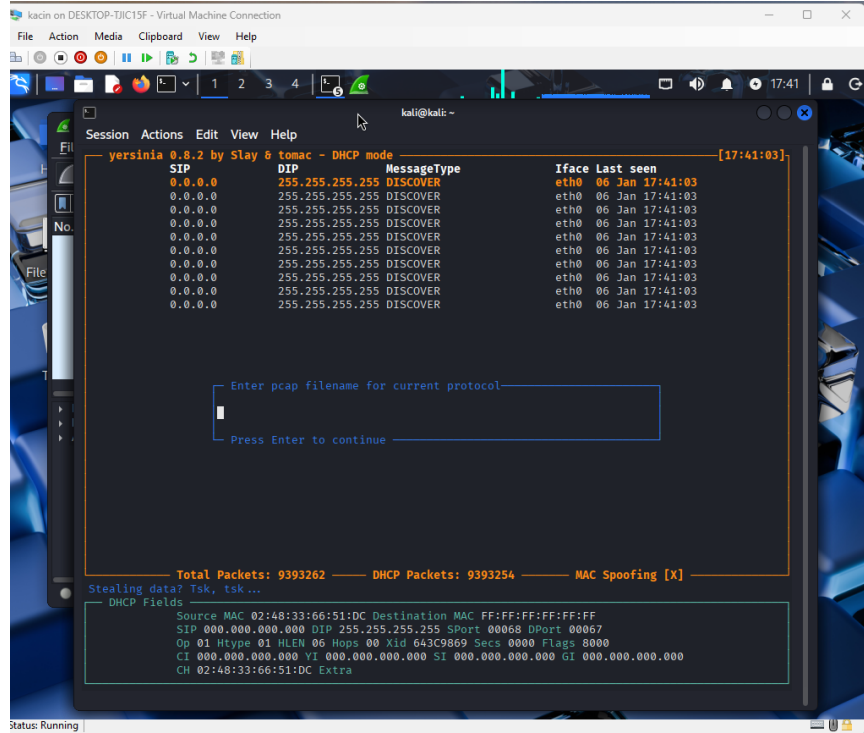
Mac Address Table			
Vlan	Mac Address	Type	Ports
All	0100.0ccc.cccc	STATIC	CPU
All	0100.0ccc.cccd	STATIC	CPU
All	0180.c200.0000	STATIC	CPU
All	0180.c200.0001	STATIC	CPU
All	0180.c200.0002	STATIC	CPU
All	0180.c200.0003	STATIC	CPU
All	0180.c200.0004	STATIC	CPU
All	0180.c200.0005	STATIC	CPU
All	0180.c200.0006	STATIC	CPU
All	0180.c200.0007	STATIC	CPU
All	0180.c200.0008	STATIC	CPU
All	0180.c200.0009	STATIC	CPU
All	0180.c200.000a	STATIC	CPU
All	0180.c200.000b	STATIC	CPU
All	0180.c200.000c	STATIC	CPU
All	0180.c200.000d	STATIC	CPU
All	0180.c200.000e	STATIC	CPU
All	0180.c200.000f	STATIC	CPU
All	0180.c200.0010	STATIC	CPU
All	ffff.ffff.ffff	STATIC	CPU
1	b4a8.b901.b870	STATIC	Fa1/0/5
Total Mac Addresses for this criterion: 21			

Arp Spoofing:



The screenshot shows the Wireshark network protocol analyzer interface. The title bar indicates 'Capturing from eth0'. The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and analysis. The filter bar at the top shows 'icmp'. The packet list pane displays a single packet: an ICMP Echo (ping) from 192.168.1.100 to 192.168.1.1. The packet details pane shows the structure of the ICMP Echo request, including the type (0), code (0), and identifier (1). The packet bytes pane shows the raw data of the packet.

DHCP starvation:



id	address	Client-ID	Hardware address	lease expiration	Type	State	Interface
192.168.1.122	0100.1502.3801.003	Jan 07 2026	10:14 PM	Automatic	Active	GigabitEthernet0/0/0	
192.168.1.123	0100.1502.3801.004	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.124	5aee.981f.000f	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.125	0100.1502.3801.005	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.126	9630.42d2.1f58	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.127	f66c.432f.611c	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.128	a730.42d2.1f58	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.129	b45e.e972.e62c	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.130	0100.1502.3801.006	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.131	06ec.0544.5d62	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.132	c6d0.433a.b566	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.133	0100.1502.3801.007	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.134	7b11.6041.27ef	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.135	0100.1502.3801.008	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.136	0482.f4a7.1530	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.137	1049.4d42.9949	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.138	9630.42d2.1f58	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.139	ba59.874b.373c	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.140	0100.1502.3801.009	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.141	34ee.a06b.221b	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.142	0632.7472.98b7	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.143	62c7.9d55.5562	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.144	06ad.062f.9b67	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.145	0100.1502.3801.00a	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.146	7443.8761.3b76	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.147	0100.1502.3801.00b	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.148	1408.736b.ff14	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.149	a8b7.9912.020a	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.150	06b0.433a.b566	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.151	a269.4365.021f	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.152	1456.0202.0000	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.153	24fe.04ac.5b87	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.154	7490.3776.7676	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.155	947f.0202.0000	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.156	7f8b.161f.7676	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.157	0100.1502.3801.00c	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.158	909e.9454.5428	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.159	4122.42cb.6897	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.160	62f3.6334.36e2	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.161	0100.1502.3801.00d	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.162	58da.b328.4279	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.163	0100.1502.3801.00e	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.164	434f.0585.30da	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.165	6620.3600.18f1	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.166	947f.0202.0000	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.167	2aa3.760b.2b08	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.168	0100.1502.3801.00f	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.169	40e6.8e17.9369	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.170	0100.1502.3801.010	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.171	99d1.99d1.99d1	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.172	020e.0052.dffa	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.173	44f2.0202.0000	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.174	0683.575b.b668	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.175	06d3.9454.5428	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.176	0100.1502.3801.011	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.177	0640.0202.0000	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.178	16c3.9432.010b	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.179	0100.1502.3801.012	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.180	1863.913b.1b4d	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.181	0100.1502.3801.013	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.182	8241.b635.4440	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.183	3285.0202.99fa	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.184	947f.0202.0000	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.185	ea90.978e.d6d2	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.186	0100.1502.3801.014	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.187	40e6.997b.8602	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.188	1414.c10e.1b77	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.189	0100.1502.3801.015	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.190	0003.4662.c61c	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.191	0055.0202.0000	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.192	4e42.441f.8d1c	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.193	b999.9999.9999	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.194	3216.910e.09e5	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.195	265c.9023.7448	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.196	a818.0202.0000	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.197	72e5.365e.4397	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.198	0100.1502.3801.016	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	
192.168.1.199	ae41.4372.6439	Jan 07 2026	10:14 PM	Automatic	Selecting	GigabitEthernet0/0/0	

Configurations

Switch

Current configuration : 3159 bytes

```
!  
version 12.2  
no service pad  
service timestamps debug uptime  
service timestamps log uptime  
no service password-encryption  
!  
hostname Kacin  
!  
!  
no aaa new-model  
system mtu routing 1500  
ip subnet-zero  
!  
ip dhcp snooping vlan 220  
ip arp inspection vlan 220  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
!  
!  
!  
vlan internal allocation policy ascending  
!  
!  
!  
!  
interface FastEthernet1/0/1  
switchport access vlan 22  
switchport mode access  
shutdown  
!  
interface FastEthernet1/0/2  
switchport access vlan 22  
switchport mode access
```



```
shutdown
!
interface FastEthernet1/0/3
switchport access vlan 220
switchport mode access
switchport port-security maximum 3
switchport port-security violation restrict
!
interface FastEthernet1/0/4
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/5
switchport access vlan 911
switchport mode access
!
interface FastEthernet1/0/6
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/7
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/8
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/9
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/10
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/11
switchport access vlan 22
```

```
switchport mode access
shutdown
!
interface FastEthernet1/0/12
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/13
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/14
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/15
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/16
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/17
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/18
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/19
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/20
```

```
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/21
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/22
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/23
switchport access vlan 22
switchport mode access
shutdown
!
interface FastEthernet1/0/24
switchport access vlan 22
switchport mode access
shutdown
!
interface GigabitEthernet1/0/1
switchport access vlan 22
switchport mode access
shutdown
!
interface GigabitEthernet1/0/2
switchport access vlan 22
switchport mode access
shutdown
!
interface GigabitEthernet1/1/1
switchport access vlan 22
switchport mode access
shutdown
!
interface GigabitEthernet1/1/2
switchport access vlan 22
switchport mode access
shutdown
!
```

```
interface Vlan1
no ip address
!
ip classless
!
!
ip http server
ip http secure-server
!
!
!
control-plane
!
!
line con 0
line vty 5 15
!
end
```

Problems

To successfully complete the lab, I first needed to find virtualization software, but because I installed a version of windows with Hyper-V preinstalled it came with security features regarding VM's that caused other virtualization software to not function as intended. I initially tried to use VMware and VirtualBox as they are commonly used, but none of them worked besides Hyper-V. Once I tried using Hyper-V, I had problems configuring the Virtual Switch which is responsible for connecting the VMs to the external network, after I bound it to the proper network adapter, I was able to connect to the physical network to provide my VMs with internet to update the Kali Linux hacking tool repository.

I also had a problem with getting the updated versions of the hacking tools; to update the tools I had to let the Kali Operating System access the internet. To do this, I needed to configure the router with default routes, PAT, and as a DHCP client. This took some time but was not difficult, but when creating the default route, I did not create a fully specified route that Ethernet needs to work but rather 2 routes: one with an IP address and another route using the outgoing interface. Ethernet did not like this, and the route would not function properly; it took some time to figure out that I needed a fully specified default route and once implemented the VMs could access the internet, and I could run the hacking tools effectively.

Conclusion

This lab was extremely helpful in understanding how Layer 2 attacks work and how the mitigations actively work to prevent those attacks. This understanding of how Layer 2 vulnerabilities are exploited and the methods to mitigate them has helped me further develop my skillset in access control and Layer 2 security. By doing this lab, I can combine theory and application to further develop the extent to which my skills can apply to, such as how to prevent other attacks besides the three specifically mentioned in this lab. This lab also helped me gain familiarity and skills in Hyper-V and Linux machines which are great as these software and Operating Systems are commonly used in enterprise environments. Most of all, I enjoyed the hacking part of this lab as it was interesting to learn how penetration testers and exploiters work to break into systems and how companies and networks rebuild themselves even stronger to prevent those attacks.